

Answer for Practice Questions II

Economics 142

1. Choose the correct letter for each question.
 - (a) When one good is sold to one of the N buyers with value $v_1 > v_2 > \dots > v_N > 0$...
 - A. The buyer-optimal market-clearing price is v_N .
 - B. The buyer-optimal market-clearing price is v_2 .
 - C. The buyer-optimal market-clearing price is v_1 .
 - D. The buyer-optimal market-clearing price and the seller-optimal market-clearing price is the same.

Answer. B (the seller-optimal market-clearing price is v_1 .)

- (b) Consider an arbitrary bipartite network with N unit-sellers and N unit-buyers with heterogeneous values. They can trade only when they are connected. If we add one more buyer who is connected to every seller with value $v_{iN+1} > 0$ for $i = 1, \dots, N$, then....
 - A. All original N buyers must be weakly better off given the new buyer-optimal market clearing prices.
 - B. Some buyer must be strictly worse off given the new buyer-optimal market clearing prices.
 - C. The total values at the efficient assignment must increase weakly.
 - D. The total values at the efficient assignment must decrease weakly.

Answer. C (The set of feasible matchings expands with more buyers or sellers. So, it always improves efficiency to add more buyers or sellers.)

- (c) Suppose that θ is either 1 and -1 with equal probability and let $s \in \{s_1, \dots, s_m\}$ be a signal about θ , which follows some conditional distribution $\Pr[s = s_i | \theta = z], z = 1, -1$.

- A. It is possible that the posterior belief $\Pr[\theta = 1 | s = s_i]$ for $\theta = 1$ is strictly larger than 0.5 for every $i = 1, \dots, m$.
- B. It is possible that the posterior belief $\Pr[\theta = 1 | s = s_i]$ is strictly strictly smaller than 0.5 for every $i = 1, \dots, m$.
- C. If the posterior belief $\Pr[\theta = 1 | s = s_i]$ is strictly larger than 0.5 for some s_i , then it must be strictly smaller than 0.5 given some other $s_j \neq s_i$.
- D. The expected posterior belief $E[\Pr[\theta = 1 | s = s_i]] = \sum_{i=1}^m \Pr[\theta = 1 | s = s_i] \Pr[s = s_i]$ can be different from 0.5.

Answer. C (This follows from the fact that the expected posterior belief must be equal to the prior belief (slide P.8))

- (d) Consider a star network with $n + 1$ nodes. Assume that each node i observes a noisy signal $x_i(0) = \theta + \epsilon_i$ of some hidden state $\theta \in \mathbb{R}$, where ϵ_i is i.i.d. noise with mean 0. Let 0 be the center node. Assume that every node updates its x by taking the averages of x of its own and its neighbors ($x_0(t+1) = \frac{\sum_{i=0}^n x_i(t)}{n+1}$ for the center node and $x_i(t+1) = \frac{x_0(t) + x_i(t)}{2}$ for other $i \neq 0$). Let $x_n(\infty)$ be the consensus that is reached as $t \rightarrow \infty$.

- A. Given each θ , $x_n(\infty)$ converges to some number as $n \rightarrow \infty$ and may be biased (i.e. $\lim_{n \rightarrow \infty} x_n(\infty) \neq \theta$).
- B. Given each θ , $x_n(\infty)$ converges to θ .
- C. Given each θ , $x_n(\infty)$ is random even as $n \rightarrow \infty$ and may be biased (i.e. $E[\lim_{n \rightarrow \infty} x_n(\infty)] \neq \theta$).
- D. Given each θ , $x_n(\infty)$ is random even as $n \rightarrow \infty$ and is unbiased (i.e. $E[\lim_{n \rightarrow \infty} x_n(\infty)] = \theta$).

Answer. D

(Remember that $x_n(\infty) = \sum_{i=0}^n w_i^* x_i(0) = w_c^* x_1(0) + n w_p^* (\sum_{i=1}^n x_i(0))$, where w_c^* is the influence weight for the center node and w_p^* is the influence weight for each peripheral node. We can find $w_c^* = \frac{n+1}{3n+1}$

and $w_p^* = \frac{2}{3n+1}$ by solving the system of equations like the one in P. 19 of the slide (here we have $n+1$ nodes instead of n nodes). Hence $x_n(\infty) = \frac{n+1}{3n+1}x_1(0) + \frac{2}{3n+1}(\sum_{i=1}^n x_i(0))$. This remains random given each θ even when $n \rightarrow \infty$ and it is unbiased as the expectation of each $x_i(0)$ is the true state θ .)

2. Various types of memberships for some subscription service are on sale. Suppose that currently there is only one available slot of a special membership with service level x_s , then there are two slots for regular memberships of service level $x_r \in (0, x_s)$. Suppose that three people $j = 1, 2, 3$ are interested in these three memberships. Assume that j values service x by $\theta_j x$, where $\theta_1 > \theta_2 > \theta_3 > 0$.

Answer the following questions.

- (a) What is the most efficient assignment/allocation of the three memberships to these three people?
- (b) Find the smallest (i.e. buyer-optimal) market-clearing prices.
- (c) Find the largest (i.e. seller-optimal) market-clearing prices.
- (d) Suppose that one regular membership is replaced with a upgraded membership “regular+”. The service x_{r+} for a regular+ membership is between x_r and x_s . Find the buyer-optimal market-clearing prices and the seller-optimal market clearing prices and compare them to your answer for (c). (We assume that one regular membership is withdrawn so that there are three slots of memberships. As an exercise, do the same question assuming that two regular memberships are still available (so there are four membership slots available)).

Answer.

- (a) Assign the special membership to 1, and two regular memberships to 2 and 3.

- (b) Let p_s^* be the buyer-optimal market-clearing price for a special membership and p_r^* be the buyer-optimal market clearing price for a regular membership. Given any market-clearing price, 2 and 3 must purchase a regular membership. $p_r^* = 0$ is the cheapest possible price. Since p_s^* must satisfy $\theta_2 x_r - p_r^* \geq \theta_2 x_s - p_s^*$, we have $p_s^* = \theta_2(x_s - x_r)$ (slide p.24).

(c) Let $\hat{p}_s$ be the seller-optimal market-clearing price for a special membership and $\hat{p}_r$ be the seller-optimal market clearing price for a regular membership. They satisfy $\theta_3 x_r - \hat{p}_r \geq 0$, $\theta_1 x_s - p_s \geq \theta_1 x_r - \hat{p}_r$. Hence $\hat{p}_r = \theta_3 x_r$ and $\hat{p}_s = \theta_1(x_s - x_r) + \hat{p}_r = \theta_1(x_s - x_r) + \theta_3 x_r$.

(d) Let p_{r+}^* be the buyer-optimal market-clearing price and $\hat{p}_{r+}$ be the seller-optimal market-clearing price for a regular+ membership.

- Buyer-Optimal: They satisfy $p_r^* \geq 0$, $\theta_3 x_r - p_r^* \geq \theta_3 x_{r+} - p_{r+}^*$, $\theta_2 x_{r+} - p_{r+}^* \geq \theta_2 x_s - p_s^*$. They are the lowest prices that satisfy these inequalities. Hence $p_r^* = 0$, $p_{r+}^* = \theta_3(x_{r+} - x_r)$, and $p_s^* = \theta_2(x_s - x_{r+}) + \theta_3(x_{r+} - x_r)$.
- Seller-Optimal: They satisfy $\theta_3 x_r - p_r^* \geq 0$, $\theta_2 x_{r+} - p_{r+}^* \geq \theta_2 x_r - p_r^*$, $\theta_1 x_s - p_s^* \geq \theta_1 x_{r+} - p_{r+}^*$. They are the highest prices that satisfy these inequalities. Hence $p_r^* = \theta_3 x_r$, $p_{r+}^* = \theta_2(x_{r+} - x_r) + \theta_3 x_r$, and $p_s^* = \theta_1(x_s - x_{r+}) + \theta_2(x_{r+} - x_r) + \theta_3 x_r$.

The final matching is not affected by the replacement of a regular membership with a regular+ membership. For both the buyer-optimal market-clearing prices and the seller-optimal market-clearing prices, buyer 2 pays more and buyer 1 pays less when the regular+ membership is introduced (buyer 3 pays the same price).

3. Consider the model of observational learning where people decide whether to adopt a new technology or not sequentially. Each person decides to adopt a new strategy or stick to the old technology one by one after observing some information about the new technology and observing the actions of the preceding people. There are two possible states: the new technology is pathbreaking ($\theta = P$) with probability p or a failure ($\theta = F$) with probability $1 - p$. Each person's payoff from adopting a new technology is $v + 80$ if the state is P and $v - 120$ if the state is F , where v is the payoff from sticking to the old technology. Each person i observes a (independent and identically distributed) signal $s_i \in \{pb, f\}$, which is correct with probability $2/3$ (i.e. $\Pr[s_i = pb | \theta = P] = \Pr[s_i = f | \theta = F] = 2/3$).

- (a) If the first person observes $s_1 = f$, what is her posterior belief for $\theta = P$ and $\theta = F$?

- (b) For some range of initial belief p , everyone ignores her own signal and chooses the same action. Characterize such range of p .
- (c) Suppose that $p = 1/2$. What is the probability that the t th person for every $t \geq 3$ choose the same action independent of their own information?
- (d) When $p = 1/2$, what is the probability that a sequence of people converge to choose the same but a wrong choice?

Answer.

$$(a) \Pr[\theta = P | s_1 = f] = \frac{p \times 1/3}{p \times 1/3 + (1-p) \times 2/3}, \Pr[\theta = F | s_1 = f] = \frac{(1-p) \times 2/3}{p \times 1/3 + (1-p) \times 2/3}.$$

(b) Note that it is strictly optimal to choose the new/old technology when the posterior belief on $\theta = P$ is larger/smaller than $3/5$. If p is large, then the posterior belief is always above $3/5$ independent of private signal. Hence each private signal becomes useless. If p is small, then the posterior belief is always below $3/5$, so again everyone ignores her signal. For the former case, we can look for p such that the posterior belief is above $3/5$ even after observing f . This p can be obtained by solving $\frac{p \times 1/3}{p \times 1/3 + (1-p) \times 2/3} = \frac{3}{5}$, which is $\bar{p} = 3/4$. Similarly, we look for p such that the posterior belief is below $3/5$ even after observing pb . Hence we can solve $\frac{p \times 2/3}{p \times 2/3 + (1-p) \times 1/3} = \frac{3}{5}$ to find $\underline{p} = 3/7$. Therefore, the range of such p is $[0, 3/7]$ and $(3/4, 1]$.

(c) By (b), if $p = 1/2 \in (3/7, 3/4)$, $s_1 = pb$ would move the first person's posterior belief (on $\theta = P$) above $3/5$ (it's $2/3$) and $s_1 = f$ would move it below $3/5$ (it's $1/3$). So, the first person chooses the new technology given $s_1 = pb$ and the old technology given $s_1 = f$. For the latter case, the 2nd person can infer $s_1 = f$. In this case, even if he observes $s_2 = pb$, his posterior belief would be just back to the initial belief $1/2$, where it is optimal for him to stick to the old technology. So, the 2nd person chooses the old technology independent of his signal. For the former case, if he observes $s_2 = f$, then the posterior belief is back to $1/2$ and he chooses the old technology. Given this, the third person is in the same situation as the 1st person. If he observes $s_2 = pb$, then his posterior belief would get even larger than $2/3$. So, he would choose the new technology. Then the third person will know that two pb have been observed, so his signal would not matter any more (note that the

posterior belief given two pb and one f is the same as the posterior belief given just one pb . What matters is the difference of the numbers of realizations of pb and f). To sum up, the third person and everyone that follows him ignore their signals and take the same action when the first two signals are (pb, pb) or (f, z) for $z = pb, f$. The probability of this event is $1/2 \times ((2/3)^2 + 1/3) + 1/2((1/3)^2 + 2/3) = 7/9$.

(d) By (c), they converge to the correct action at $t = 3$ when $\theta = P$ and (pb, pb) is observed, or when $\theta = F$ and f is observed in the 1st period. When (pb, f) is observed, then they are back to the starting point at $t = 3$. The probability of the first event is $1/2 \times (2/3)^2 + 1/2 \times 2/3 = 5/9$. The probability of the second event is $1/2 \times 1/3 + 1/2 \times (1/3)^2 = 2/9$. Hence the probability of the third event is $2/9$. Whenever the convergence of actions (i.e. herding) occurs, people converge to the correct action 2.5 times more often ($5/9$ vs. $2/9$). Hence the probability to converge to the wrong action is $2/7$ (the probability of nonconvergence is 0).